

Team2 Advoteck Mobile Keyboard Extension

Team2 – Spring 2024

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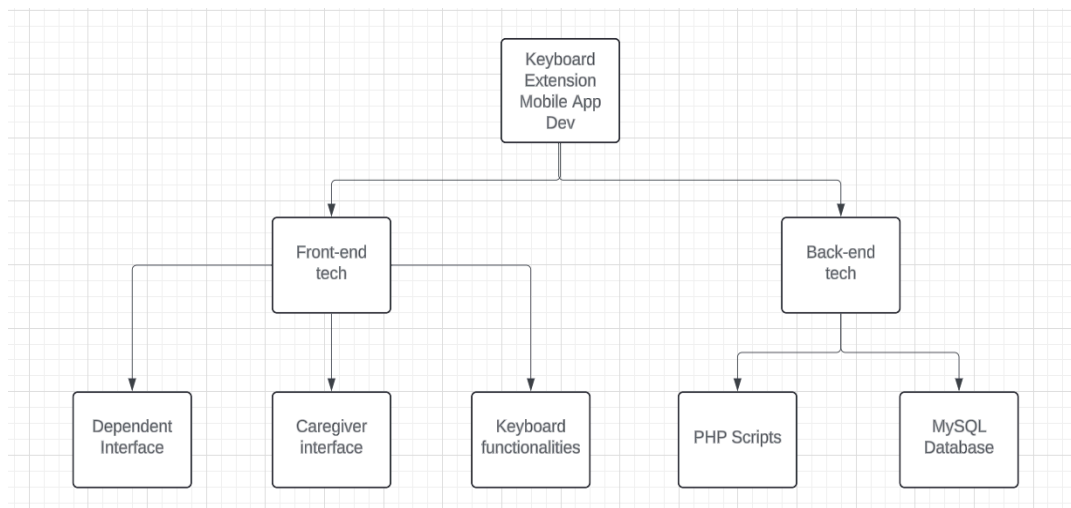
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1.0 Introduction

The Mobile Keyboard Extension is designed to enhance digital security for vulnerable users. This documentation outlines the MVP which facilitates safe information handling while maintaining a standard keyboard experience with features like next-word prediction and contextual generalization. The extension's alert system detects sensitive information such as Social Security numbers and credit card details. It warns users and, if ignored, sends redacted alerts to caregivers, ensuring timely intervention while respecting user privacy. Key features include user-centric design, robust security, comprehensive accessibility, seamless compatibility with messaging platforms, easy installation, and flexible payment options. This guide details the extension's functionalities and the strategic design ensuring user safety and caregiver readiness.

2.0 High-Level Hierarchy

2.1 Hierarchy Diagram



2.2 Hierarchy Description

The Mobile Keyboard Extension is structured to ensure digital safety for vulnerable users through a layered system architecture:

2.2.1. Mobile App (Keyboard Extension)

It serves as the main interaction point, functioning like a standard keyboard but with enhanced security features to monitor sensitive data.

2.2.2. Frontend (User Interface Components)

Dependent Interface (Android UI): Tailored for vulnerable individuals, featuring accessibility tools like Notepad.

Caregiver Interface (Android UI): Provides monitoring tools and alerts for caregivers, ensuring timely intervention with minimal information to maintain privacy.

Keyboard Functionalities (Java, XML): Core features built using Java for logic and XML for layouts, enabling real-time detection and alerting on sensitive information.

2.2.3. Backend (Server-side Components)

PHP Scripts: Handle user authentication, data validation, and processing, ensuring secure and efficient interactions between mobile interfaces and the server.

MySQL Database: Stores user data and alert system logs securely, supporting quick data retrieval for real-time alerting and historical analysis.

2.2.4. System Integration and Data Flow

Integration ensures seamless operation across the system. Inputs are processed in real time, with sensitive information detection triggering alerts. PHP scripts manage these alerts, which determine the information sent to caregivers. All data transactions are securely logged in the MySQL database, ensuring privacy and compliance with security standards. This architecture ensures robust, scalable, and secure operations, ready for future enhancements.

3.0 Components Classification

3.1 Mobile App (Android Keyboard Extension)

Purpose: The mobile app layer serves as the primary interface for end-users, integrating the keyboard extension directly into the environment. It is designed to enhance digital safety by providing a secure, intuitive keyboard that detects sensitive information as it is typed.

Specific Nature: This layer is developed specifically for Android, using technologies compatible with Android development standards to ensure seamless integration and operation. The app replicates the look and feel of native Android keyboard applications while adding advanced security features.

Subcomponents and Associated Constructs:

- **User Authentication:** Ensures that each session is securely initiated by authorized users only.

- **Settings Management:** Allows users to customize settings such as sensitivity levels of information detection and specific data to be monitored.
- **Interface Feedback:** Provides real-time feedback to users when sensitive information is detected, using Android native alerts and notifications.

3.2 Frontend (User Interface Components)

Purpose: The front end focuses on providing a user-friendly experience, enabling different users (dependents and caregivers) to interact with the application effectively. It ensures that the application is accessible, intuitive, and easy to navigate.

Specific Nature: Developed using Android UI principles adapted for Android standards, the frontend layer is responsible for rendering user interfaces on the device screen, capturing user inputs, and displaying appropriate responses based on backend processing.

Subcomponents and Associated Constructs:

- **Dependent Interface:** Tailored for vulnerable users, featuring large buttons, readable text, and support for accessibility features.
- **Caregiver Interface:** Offers a dashboard-like environment where caregivers can receive alerts, monitor dependent activities, and manage intervention settings.
- **Keyboard Functionalities:** Built using Java for backend logic and XML for frontend layouts, this component includes algorithms for sensitive information detection, sensitive word prediction, and contextual generalization, enhancing the predictive capabilities of the keyboard.

3.3 Backend (Server-side Components)

Purpose: The backend layer manages all server-side logic, including data processing, alert generation, and communication between the mobile interface and the database. It acts as the central hub for application logic and data management.

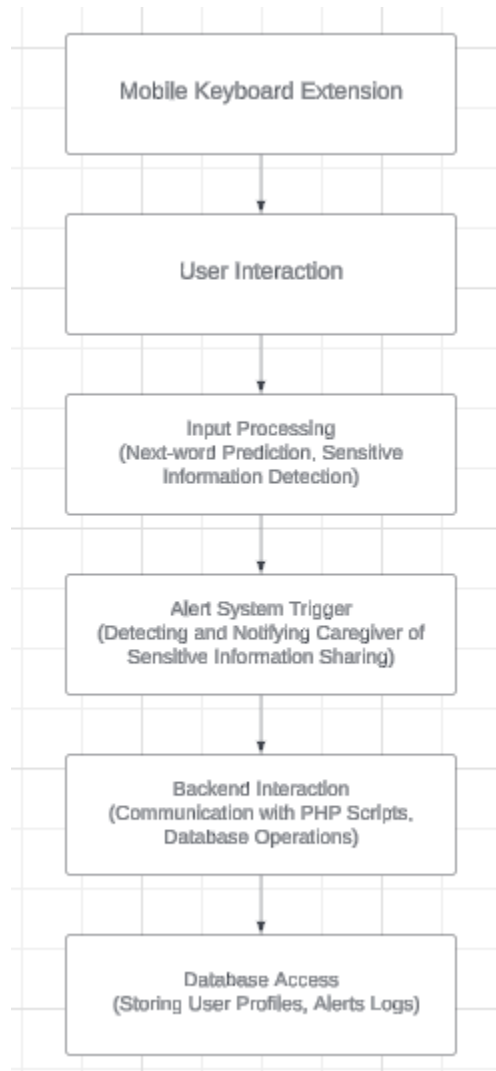
Specific Nature: Composed of server-side scripts and a database management system, the backend ensures that all data is processed securely and efficiently, maintaining integrity and privacy.

Subcomponents and Associated Constructs:

- **PHP Scripts:** Handle complex operations such as user authentication, data validation, processing of sensitive information, and generating alerts. They ensure that the system's responses are timely and accurate.
- **MySQL Database:** Stores all necessary data including user profiles, sensitivity settings, logs of detected information, and audit trails. It is structured to facilitate quick data retrieval for real-time operations while ensuring data security through encrypted storage solutions.

Each layer in this architecture is designed to fulfill specific roles while working in harmony to provide a comprehensive solution for digital safety through the mobile keyboard extension. This layered approach not only enhances security and usability but also ensures that the system can evolve with changing requirements and technological advancements.

4.0 Process View



4.1 Process View Description

The Process View is essential in understanding how the separate components and subcomponents communicate with each other in a concurrent application. By better understanding the necessary paths of communication between the components, it may be

possible to optimize the data flow and storage of the application, as well as ensuring thread-safety.

4.1.1. User Interaction (Typing on Keyboard):

User interaction represents the initial action of the dependent user typing on the mobile keyboard extension. This layer captures user input, including text and commands, initiating the process of input processing.

4.1.2. Input Processing (Next-word Prediction, Sensitive Information Detection):

Input processing involves analyzing user input to provide next-word predictions and detect potentially sensitive information. This layer utilizes algorithms to predict the next word accurately and identify predefined patterns indicative of sensitive data (e.g., Social Security numbers, addresses).

4.1.3. Alert System Trigger (Detecting and Notifying Caregiver of Sensitive Information Sharing):

The alert system trigger layer activates upon detecting potential sharing of sensitive information by the dependent user. When sensitive information is detected, the system triggers a notification to the caregiver, ensuring prompt intervention while respecting user autonomy.

4.1.4. Backend Interaction (Communication with PHP Scripts, Database Operations):

Backend interaction encompasses the communication between the frontend (mobile app) and backend components. This layer involves executing PHP scripts for data processing, interacting with the MySQL database to store and retrieve user profiles, alert logs, and other relevant data.

4.1.5. Database Access (Storing User Profiles, Alerts Logs):

Database access involves storing and managing user profiles, caregiver-dependent relationships, and logs related to alert system activities. This layer ensures efficient data storage, retrieval, and management to support the functionalities of the mobile keyboard extension.